



## Analyzing Pet Adoption Trends and Factors - Gaby Castellanos

Animals shelters are currently facing an overcrowding crisis as many pets are being surrounded to animal shelters but not enough people are adopting. This leads some to resort to controversial methods to reduce overcrowding. It's often unclear which pet characteristics and circumstances are most influential on whether a pet is adopted or not. This project aims to analyze pet shelter data to identify the most common key factors that affect the likelihood of a pet being adopted. The factors relating to a pet's amount of time spent in a shelter will also be examined. The variables included in the dataset are PetID, PetType, Breed, AgeMonths, Color, Size, Weightkg, Vaccinated, HealthCondition, TimeInShelterDays, AdoptionFee, PreviousOwner, and AdoptionLikelihood. With the exploration and analysis of the dataset, hopefully the key factors will reveal what is needed to improve adoption outcomes. This project will be done individually.

```
In [112... # Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force\_remount=True).

```
In [113... # Import packages, load the pet_adoption_data dataset, and view the data
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, confusion_matrix

PetAdoptionData = pd.read_csv('/content/drive/My Drive/Colab Notebooks/pet_adc
PetAdoptionData.head()
```

```
Out[113... 
```

|          | <b>PetID</b> | <b>PetType</b> | <b>Breed</b>     | <b>AgeMonths</b> | <b>Color</b> | <b>Size</b> | <b>WeightKg</b> | <b>Vaccinated</b> |
|----------|--------------|----------------|------------------|------------------|--------------|-------------|-----------------|-------------------|
| <b>0</b> | 500          | Bird           | Parakeet         | 131              | Orange       | Large       | 5.039768        | 1                 |
| <b>1</b> | 501          | Rabbit         | Rabbit           | 73               | White        | Large       | 16.086727       | 0                 |
| <b>2</b> | 502          | Dog            | Golden Retriever | 136              | Orange       | Medium      | 2.076286        | 0                 |
| <b>3</b> | 503          | Bird           | Parakeet         | 97               | White        | Small       | 3.339423        | 0                 |
| <b>4</b> | 504          | Rabbit         | Rabbit           | 123              | Gray         | Large       | 20.498100       | 0                 |

```
In [114... # Look over the data information and descriptive statistics
PetAdoptionData.info()
PetAdoptionData.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2007 entries, 0 to 2006
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   PetID                  2007 non-null  int64
1   PetType                2007 non-null  object
2   Breed                  2007 non-null  object
3   AgeMonths              2007 non-null  int64
4   Color                  2007 non-null  object
5   Size                   2007 non-null  object
6   WeightKg               2007 non-null  float64
7   Vaccinated             2007 non-null  int64
8   HealthCondition        2007 non-null  int64
9   TimeInShelterDays      2007 non-null  int64
10  AdoptionFee            2007 non-null  int64
11  PreviousOwner          2007 non-null  int64
12  AdoptionLikelihood     2007 non-null  int64
dtypes: float64(1), int64(8), object(4)
memory usage: 204.0+ KB
```

```
Out[114...      PetID  AgeMonths  WeightKg  Vaccinated  HealthCondition  Tin
count  2007.000000  2007.000000  2007.000000  2007.000000  2007.000000
mean    1503.000000    92.279522    15.705776     0.701046     0.196313
std     579.515315    52.148363     8.327749     0.457914     0.397307
min     500.000000     1.000000     1.018198     0.000000     0.000000
25%    1001.500000    48.000000     8.730396     0.000000     0.000000
50%    1503.000000    94.000000    15.925416     1.000000     0.000000
75%    2004.500000   138.000000    22.737180     1.000000     0.000000
max    2506.000000   179.000000    29.995628     1.000000     1.000000
```

```
In [115... # Check for missing values
PetAdoptionData.isnull().sum()
```

Out[115...

|                           |          |
|---------------------------|----------|
|                           | <b>0</b> |
| <b>PetID</b>              | 0        |
| <b>PetType</b>            | 0        |
| <b>Breed</b>              | 0        |
| <b>AgeMonths</b>          | 0        |
| <b>Color</b>              | 0        |
| <b>Size</b>               | 0        |
| <b>WeightKg</b>           | 0        |
| <b>Vaccinated</b>         | 0        |
| <b>HealthCondition</b>    | 0        |
| <b>TimeInShelterDays</b>  | 0        |
| <b>AdoptionFee</b>        | 0        |
| <b>PreviousOwner</b>      | 0        |
| <b>AdoptionLikelihood</b> | 0        |

**dtype:** int64

There are no missing values in any of the columns.

In [116... *# Check data types*  
PetAdoptionData.dtypes

Out[116...

0

|                           |         |
|---------------------------|---------|
| <b>PetID</b>              | int64   |
| <b>PetType</b>            | object  |
| <b>Breed</b>              | object  |
| <b>AgeMonths</b>          | int64   |
| <b>Color</b>              | object  |
| <b>Size</b>               | object  |
| <b>WeightKg</b>           | float64 |
| <b>Vaccinated</b>         | int64   |
| <b>HealthCondition</b>    | int64   |
| <b>TimeInShelterDays</b>  | int64   |
| <b>AdoptionFee</b>        | int64   |
| <b>PreviousOwner</b>      | int64   |
| <b>AdoptionLikelihood</b> | int64   |

**dtype:** object

In [117...

```
# Check for outliers
Q1 = PetAdoptionData['TimeInShelterDays'].quantile(0.25)
Q3 = PetAdoptionData['TimeInShelterDays'].quantile(0.75)
IQR = Q3 - Q1

outliers = PetAdoptionData[(((PetAdoptionData['TimeInShelterDays']
                               < (Q1 - 1.5 * IQR)) |
                               (PetAdoptionData['TimeInShelterDays'] >
                                (Q3 + 1.5 * IQR))))]

outliers
```

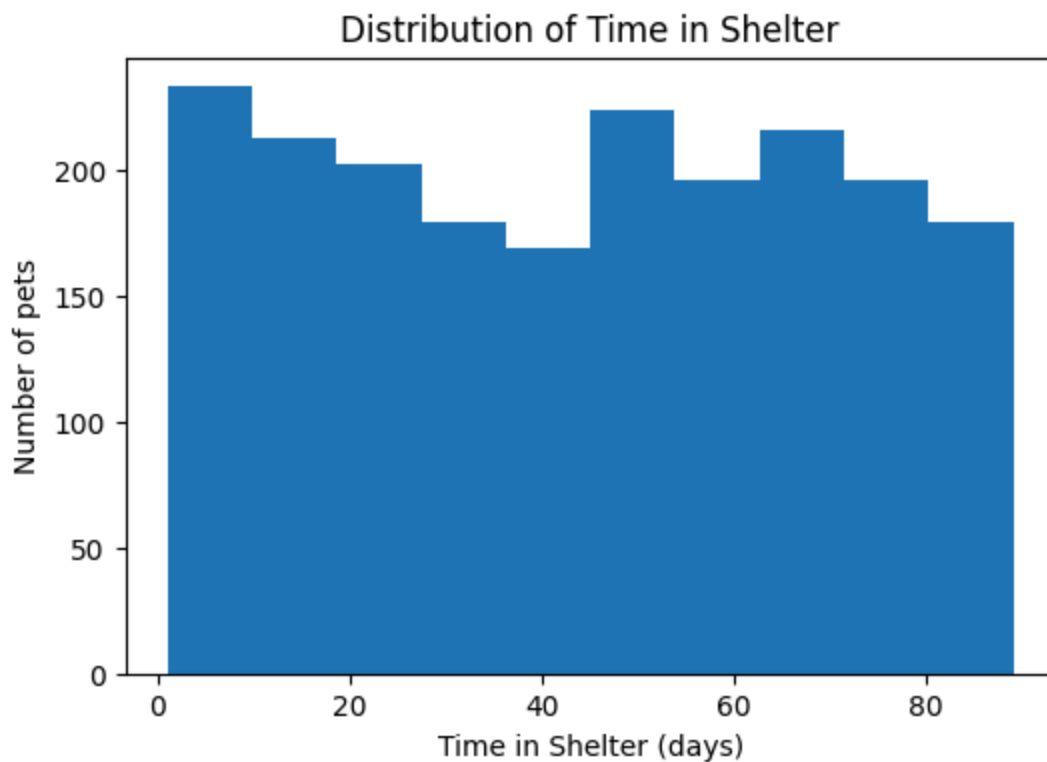
Out[117...

| PetID | PetType | Breed | AgeMonths | Color | Size | WeightKg | Vaccinated | Health |
|-------|---------|-------|-----------|-------|------|----------|------------|--------|
|-------|---------|-------|-----------|-------|------|----------|------------|--------|

There seems to be no outliers for any category when using the quantile method.

In [118...

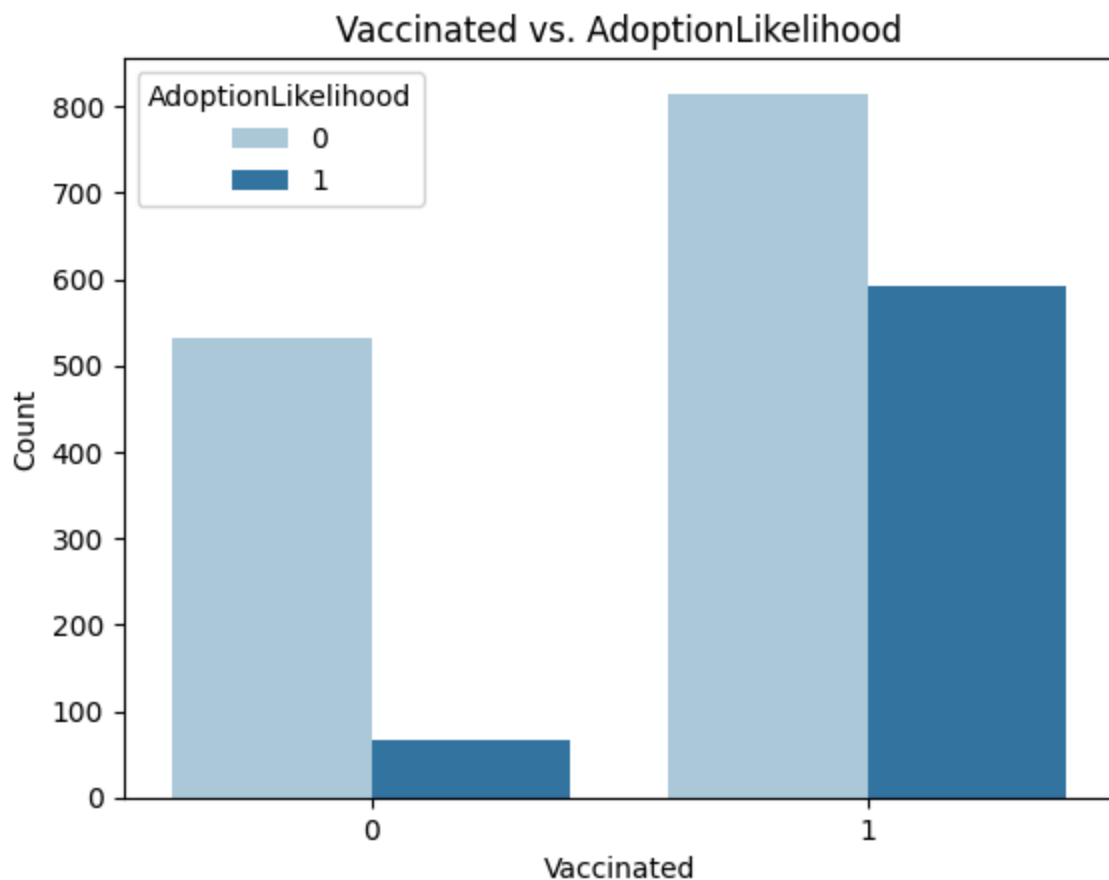
```
# Visualization of the TimeInShelterDays
plt.figure(figsize=(6, 4))
plt.hist(PetAdoptionData['TimeInShelterDays'])
plt.xlabel('Time in Shelter (days)')
plt.ylabel('Number of pets')
plt.title('Distribution of Time in Shelter')
plt.show()
```



The distribution of time in shelter appears to be fairly distributed, which suggests that longer and shorter stays occur with similar frequency. Adoption time may depend on a combination of factors.

```
In [119... # Vaccinated vs. AdoptionLikelihood
sns.countplot(x='Vaccinated', hue='AdoptionLikelihood',
              data=PetAdoptionData, palette='Paired')
plt.xlabel('Vaccinated')
plt.ylabel('Count')
plt.title('Vaccinated vs. AdoptionLikelihood')
```

```
Out[119... Text(0.5, 1.0, 'Vaccinated vs. AdoptionLikelihood')
```

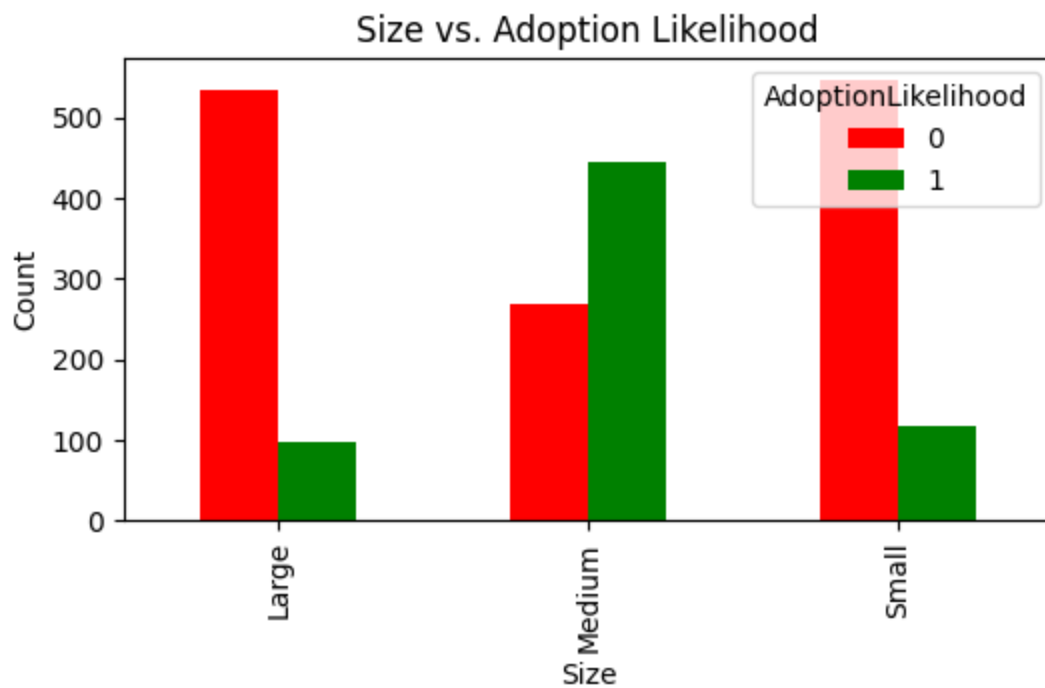


The count plot shows that vaccinated pets tend to have a higher number of successful adoptions compared to unvaccinated pets. While both groups include a high number of pets that weren't adopted, the proportion of adopted pets is noticeably higher among those that are vaccinated.

```
In [120... # Size vs Adoption Likelihood
PetAdoptionData.groupby(
    ['Size', 'AdoptionLikelihood']).size().unstack().plot(kind="bar"
    , color=['red', 'green'], figsize=(6, 3))

plt.xlabel('Size')
plt.ylabel('Count')
plt.title('Size vs. Adoption Likelihood')
```

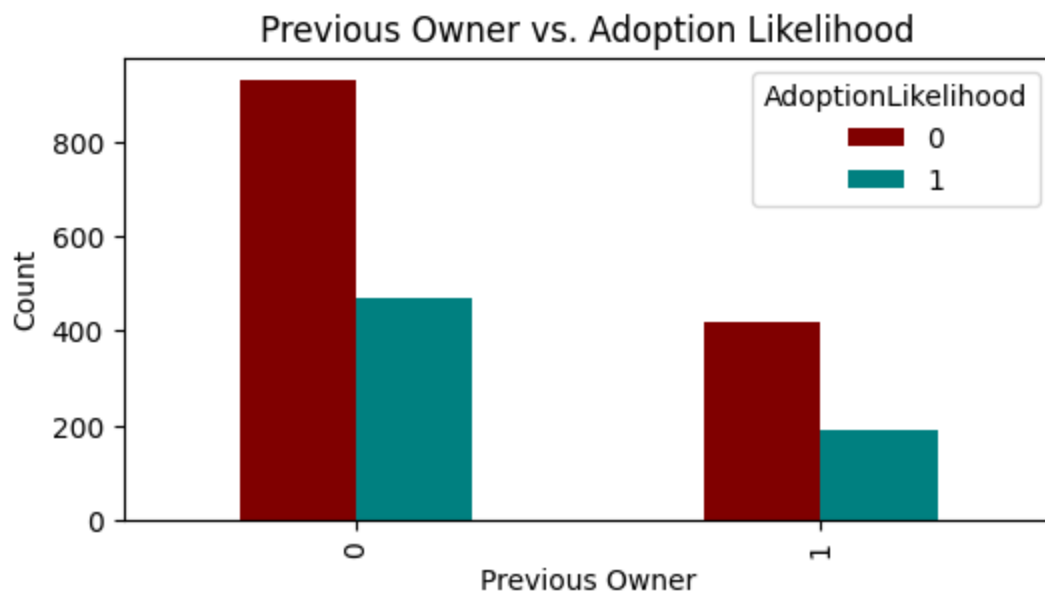
```
Out[120... Text(0.5, 1.0, 'Size vs. Adoption Likelihood')
```



There is a high number of large and small pets that don't get adopted while there is a lower number for medium pets. They also have the highest adoption numbers. This suggests that medium sized pets are the most desirable size for pets.

```
In [121]: # Previous Owner vs Adoption Likelihood
PetAdoptionData.groupby(
    ['PreviousOwner', 'AdoptionLikelihood']).size().unstack().plot(kind="bar",
    color=['maroon', 'teal'], figsize=(6, 3))
plt.xlabel('Previous Owner')
plt.ylabel('Count')
plt.title('Previous Owner vs. Adoption Likelihood')
```

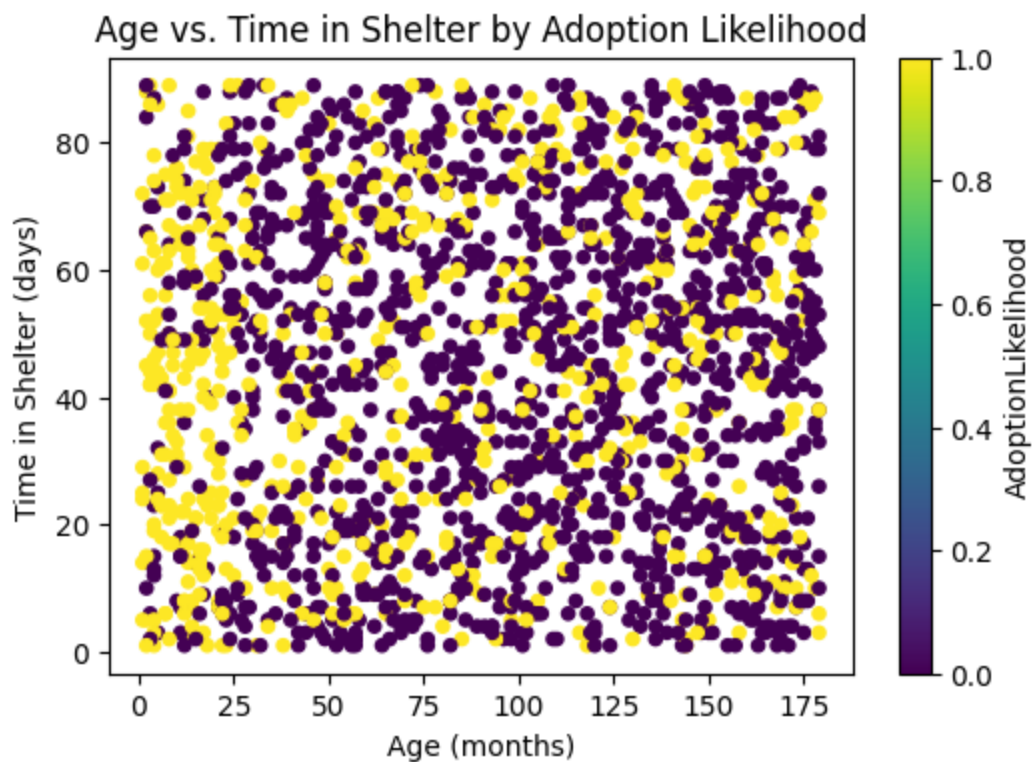
```
Out[121]: Text(0.5, 1.0, 'Previous Owner vs. Adoption Likelihood')
```



The bar chart shows that an extremely high number of pets that don't have previous owners are not adopted, which is a surprise. However, out of the pets that do get adopted, not having a previous owner seems to be favorable.

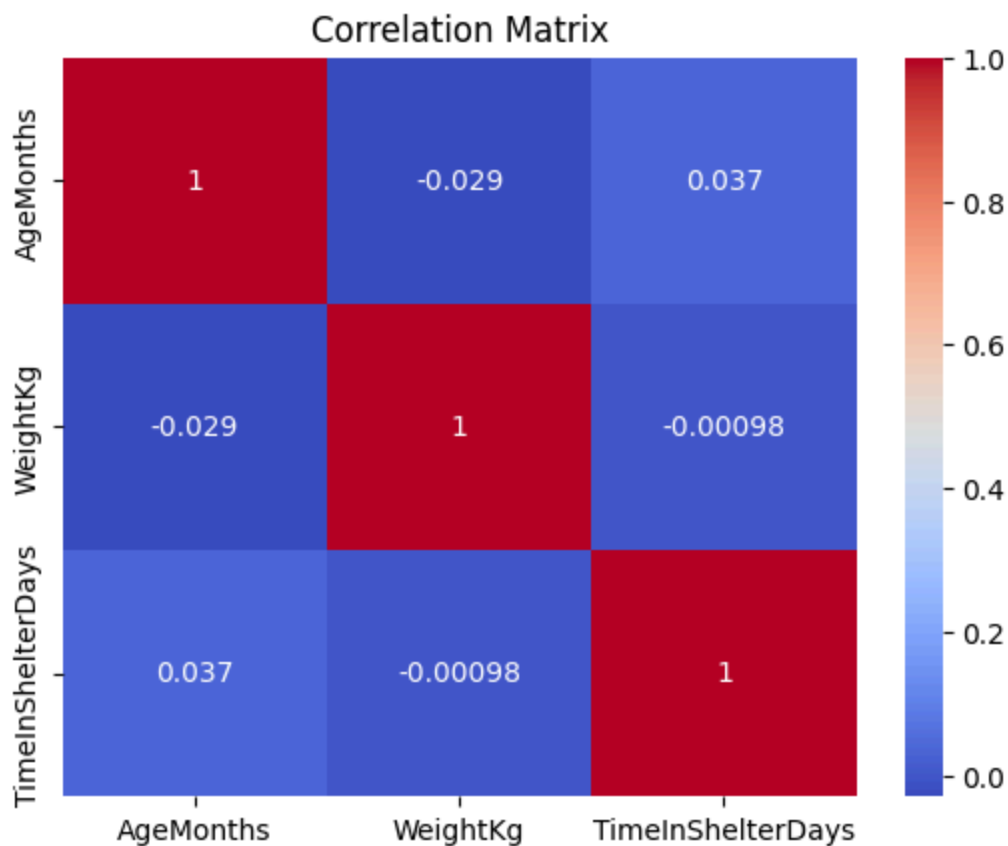
```
In [122... # Age vs Time in Shelter by Adoption Likelihood
PetAdoptionData.plot(kind='scatter', x='AgeMonths', y='TimeInShelterDays',
                      c='AdoptionLikelihood', colormap='viridis', figsize=(6, 4))
plt.xlabel('Age (months)')
plt.ylabel('Time in Shelter (days)')
plt.title('Age vs. Time in Shelter by Adoption Likelihood')
plt.show()
```





The scatter plot shows almost no correlation between a pet's age and their time in the shelter. However, the yellow dots being slightly more visible on the left side of the scatter plot may suggest that pets under 25 months old have a higher adoption likelihood.

```
In [123... # Correlation Matrix
sns.heatmap(PetAdoptionData[['AgeMonths', 'WeightKg',
                             'TimeInShelterDays']].corr(),
            annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```



The correlation matrix shows the relationships between a pet's age, weight, and time in the shelter. Since all the values are close to 0, there seems to be a weak relationship between them. Overall, this may suggest that the numerical features alone aren't strong predictors of adoption outcomes or shelter time duration.

```
In [124... # Create a copy of PetAdoptionData to use for modeling
PetAdoptionData_copy = PetAdoptionData.copy()

In [125... # Drop ID column because it is not needed for prediction
PAD = PetAdoptionData_copy.drop(columns=['PetID'], errors="ignore")

In [126... # Convert categorical variable into numeric variables.
# This makes it easier to build a model
PetAdoptionData_encoded = pd.get_dummies(PAD, drop_first=True)
PetAdoptionData_encoded.head()
```

Out[126...

|   | AgeMonths | WeightKg  | Vaccinated | HealthCondition | TimeInShelterDays | Ado |
|---|-----------|-----------|------------|-----------------|-------------------|-----|
| 0 | 131       | 5.039768  | 1          | 0               | 27                |     |
| 1 | 73        | 16.086727 | 0          | 0               | 8                 |     |
| 2 | 136       | 2.076286  | 0          | 0               | 85                |     |
| 3 | 97        | 3.339423  | 0          | 0               | 61                |     |
| 4 | 123       | 20.498100 | 0          | 0               | 28                |     |

5 rows × 23 columns

In [127...

```
# Define features and target

# AdoptionLikelihood is removed because it is what I am trying
# to predict
X = PetAdoptionData_encoded.drop(columns=['AdoptionLikelihood'])

# AdoptionLikelihood is what the model is learning to predict
y = PetAdoptionData_encoded['AdoptionLikelihood']
```

In [128...

```
# Split data into training data used to build the model
# and Testing data used to evaluate the model

# 20% of the data is used for testing while the rest is used
# for training
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size=0.2,
                                                    random_state=42)
```

In [129...

```
# Scale the data

# Puts the numeric features on a similar scale, which helps
# the logistic regression run more smoothly
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

In [130...

```
# Train the model

# Logistic regression is used as a classification model to
# predict AdoptionLikelihood
model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)
```

Out[130...

▼ **LogisticRegression** ⓘ ?

LogisticRegression(max\_iter=5000)

```
In [131... # Model accuracy
y_pred = model.predict(X_test)
accuracy_score(y_test, y_pred)
```

```
Out[131... 0.9079601990049752
```

The accuracy of the model is about 91%, which shows a high percentage of correct predictions. However, the accuracy shouldn't be the only thing used to see if AdoptionLikelihood was predicted well.

```
In [132... # Classification Report
print(classification_report(y_test, y_pred))
confusion_matrix(y_test, y_pred)
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.90      | 0.97   | 0.93     | 270     |
| 1            | 0.94      | 0.77   | 0.85     | 132     |
| accuracy     |           |        | 0.91     | 402     |
| macro avg    | 0.92      | 0.87   | 0.89     | 402     |
| weighted avg | 0.91      | 0.91   | 0.91     | 402     |

```
Out[132... array([[263,  7],
       [ 30, 102]])
```

The classification report gives more insight than the accuracy score. When the model predicts 0 (a lower likelihood of adoption), the precision is 91%, which shows a high accuracy. The same goes for the 89% precision for the model predicting 1 (high likelihood of adoption). The high recall percentages for 0 and 1 show that the model finds most of the actual cases, but it is definitely better at finding cases with 0. The high f1-scores show strong balances between precision and recall, but the balance is stronger with 0. For cases with 0, the model predicted 257 accurately and 13 inaccurately (false positive). For cases with 1, the model predicted 107 accurately and 25 inaccurately (false negative). Overall, the model is slightly better at predicting a pet not getting adopted compared to a pet being adopted.

```
In [137... # Highlighting important features

# The absolute values are focused on to find the strongest predictors overall
coefficients = pd.DataFrame({'Feature': X.columns,
                             'Coefficient': model.coef_[0]})

# By using AbsoluteCoefficient, whether a relationship is positive or negative
# is ignored, which helps rank the features by overall strength.
coefficients['AbsoluteCoefficient'] = coefficients['Coefficient'].abs()
coefficients = coefficients.sort_values(by='Coefficient', ascending=False)
```

```
mostImportantFeatures = coefficients.sort_values(by='AbsoluteCoefficient',
                                                ascending=False).head(10)
mostImportantFeatures
```

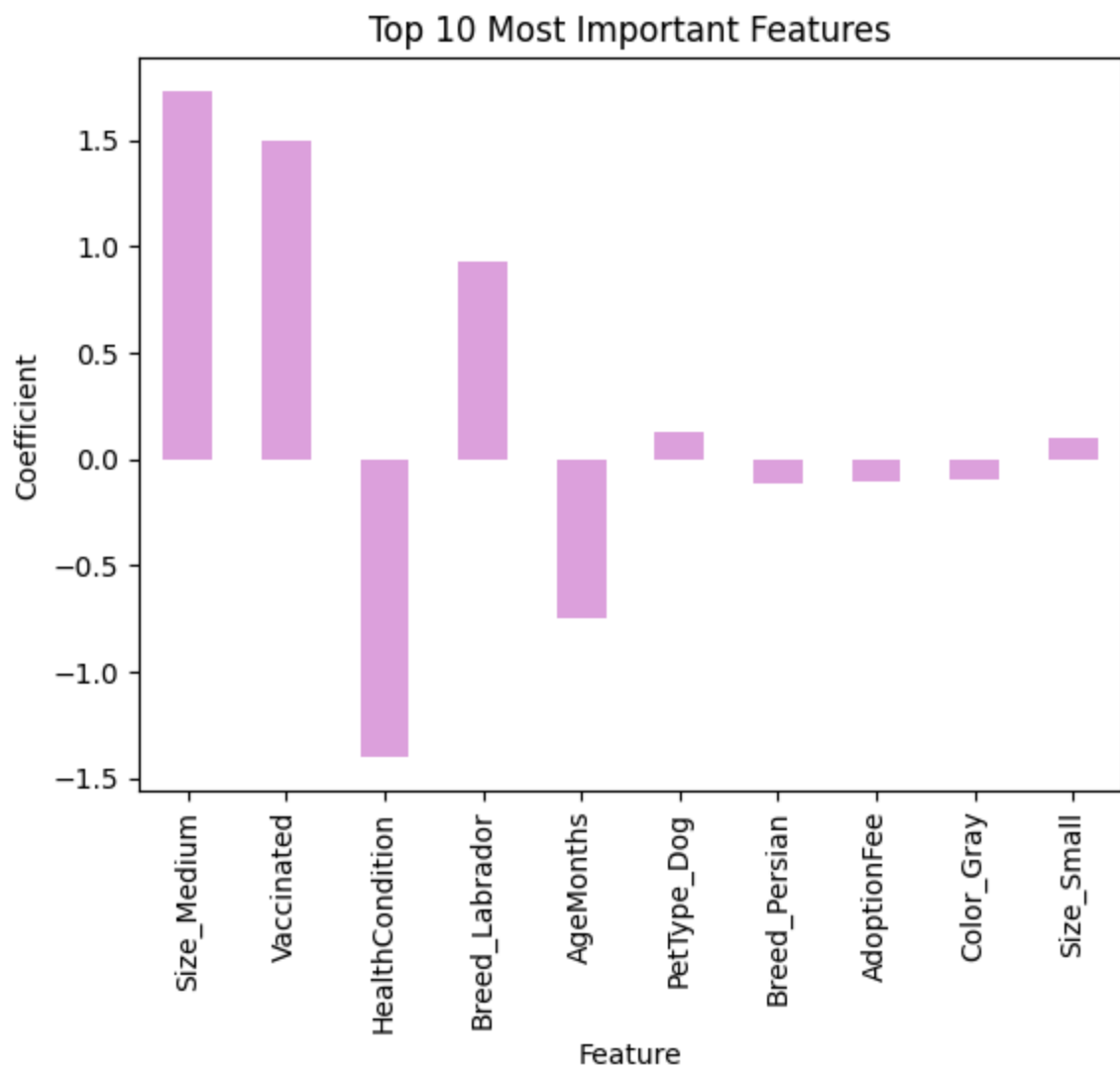
Out[137...

|           | Feature         | Coefficient | AbsoluteCoefficient |
|-----------|-----------------|-------------|---------------------|
| <b>20</b> | Size_Medium     | 1.729140    | 1.729140            |
| <b>2</b>  | Vaccinated      | 1.498223    | 1.498223            |
| <b>3</b>  | HealthCondition | -1.400746   | 1.400746            |
| <b>10</b> | Breed_Labrador  | 0.930976    | 0.930976            |
| <b>0</b>  | AgeMonths       | -0.748107   | 0.748107            |
| <b>8</b>  | PetType_Dog     | 0.129551    | 0.129551            |
| <b>12</b> | Breed_Persian   | -0.114004   | 0.114004            |
| <b>5</b>  | AdoptionFee     | -0.110220   | 0.110220            |
| <b>17</b> | Color_Gray      | -0.098036   | 0.098036            |
| <b>21</b> | Size_Small      | 0.095322    | 0.095322            |

In the chart, the positive coefficients show an increased likelihood of adoption while the negative coefficients show a decreased likelihood of adoption. The absolute coefficients show the strengths of influence of the features. The features with larger absolute coefficients have a stronger influence on pet adoptions. From the chart, a medium sized pet seems to be the most influential feature when adopting a pet, followed by vaccination status and health condition. The large positive coefficients for Size\_Medium and Vaccinated indicate that there is a preference for medium sized pets and vaccinated pets when looking at pet adoptions. The large negative coefficient for HealthCondition suggests that pets with poorer health conditions are less likely to be adopted.

In [142...

```
# Visualization of the top features
mostImportantFeatures.plot(kind='bar', x='Feature', y='Coefficient',
                           legend=False, color='plum')
plt.xlabel('Feature')
plt.ylabel('Coefficient')
plt.title('Top 10 Most Important Features')
plt.show()
```



## References

Feature Importance in Logistic Regression Models: <https://www.geeksforgeeks.org/machine-learning/understanding-feature-importance-in-logistic-regression-models/>

Classification Report and Confusion Matrix: <https://www.geeksforgeeks.org/machine-learning/compute-classification-report-and-confusion-matrix-in-python/>

Git Repo: <https://github.com/gcastellanos1/CSC108-MATH80-Repo>